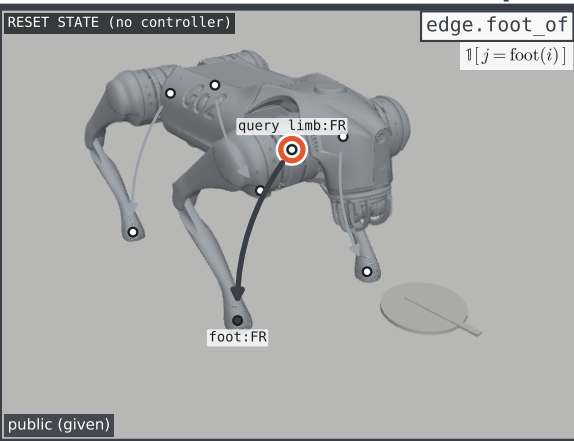
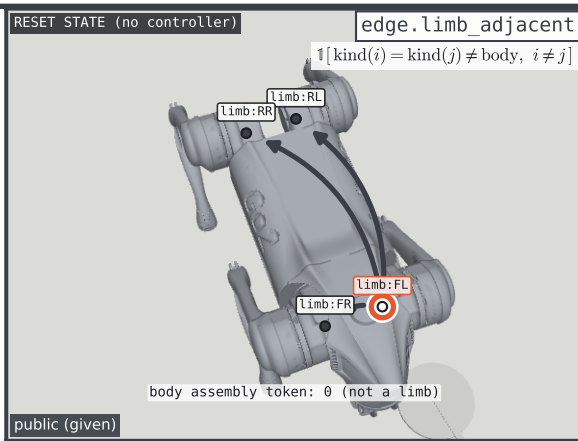


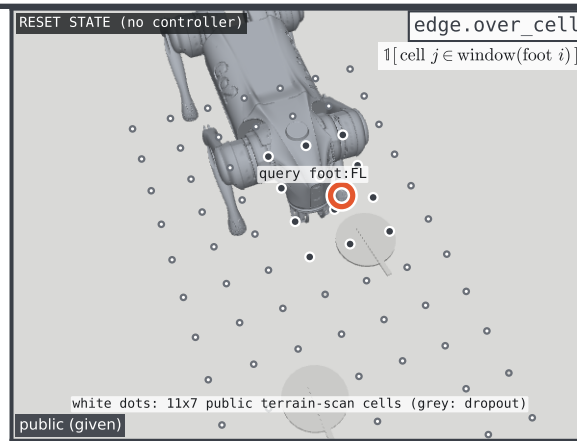
KINEMATICS / BODY (8 implemented)



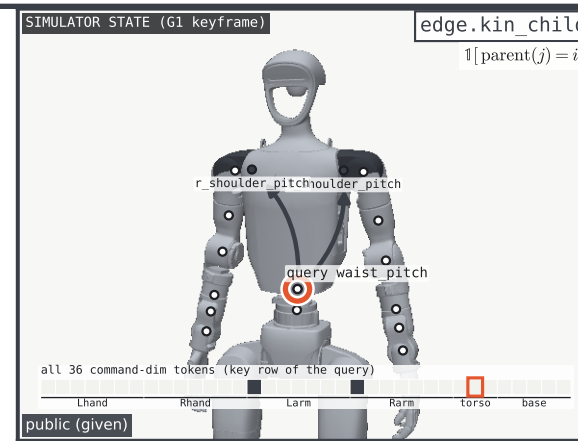
edge.foot_of: a limb token attends to its own foot token (feet exist only on foot-carrying assemblies; limbs drawn at hips, feet at foot sites). edge x bias at ctx>ctx (legged-rel-v1; w=1). Source: public morphology (LeggedMorph).



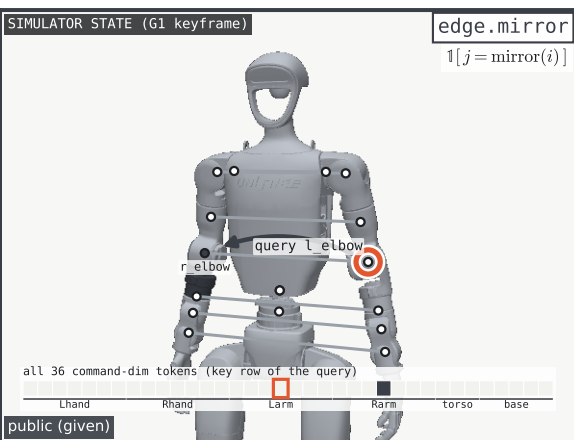
edge.limb_adjacent: a limb token attends to the other limbs of the same kind on the trunk (all four legs; tokens drawn at their hips). edge x bias at ctx>ctx (legged-rel-v1, symmetric; w=1). Source: public morphology (LeggedMorph).



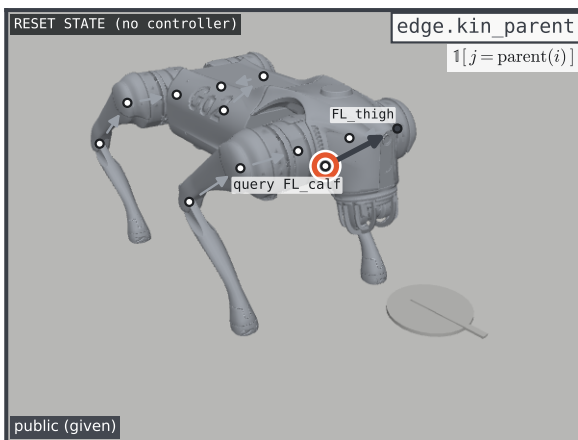
edge.over_cell: a foot token attends to the terrain-scan cells of its landing window (0.05 m behind to 0.35 m ahead, +0.15 m aside of its default stance, body frame). edge x bias at ctx>ctx (legged-rel-v1; w=1). Source: public morphology + public terrain scan.



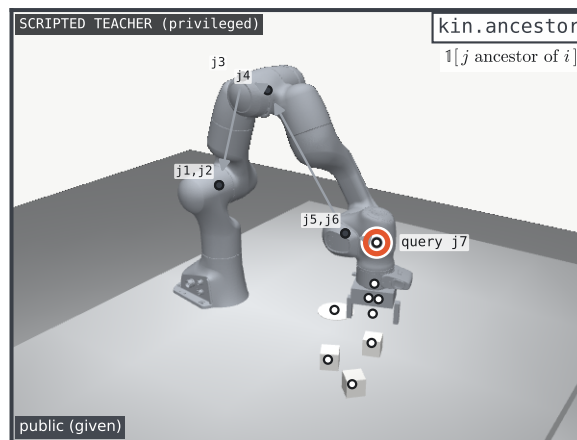
edge.kin_child: a command dim attends to its kinematic children (waist pitch -> both shoulder pitches). edge x bias at dims>dims (channel kin_child; w=1). Source: public morphology (bodies.g1_simple).



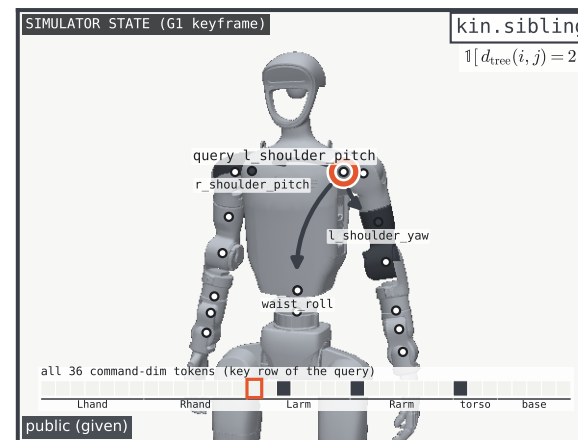
edge.mirror: a command dim attends to its left/right homologue (grey lines; every mirror pair of the body). edge x bias at dims>dims (channel mirror; w=1). Source: public morphology (bodies.g1_simple).



edge.kin_parent: a joint token attends to its kinematic parent joint (grey; every parent edge of the four legs). edge x bias at ctx>ctx (legged-rel-v1, from the per-assembly depth chain; w=1). Source: public morphology (LeggedMorph).

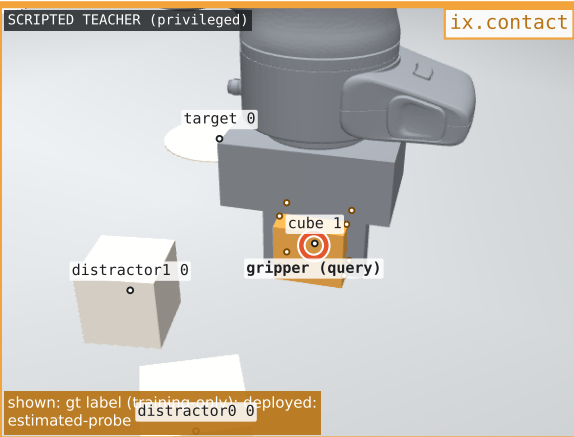


kin.ancestor: a joint token attends to every joint up its kinematic chain (transitive closure of kin_parent; grey arrows: the parent edges). ancestor (closure) x bias at ctx>ctx (arm-rel-v1; w=1). Source: public morphology (featurizer).

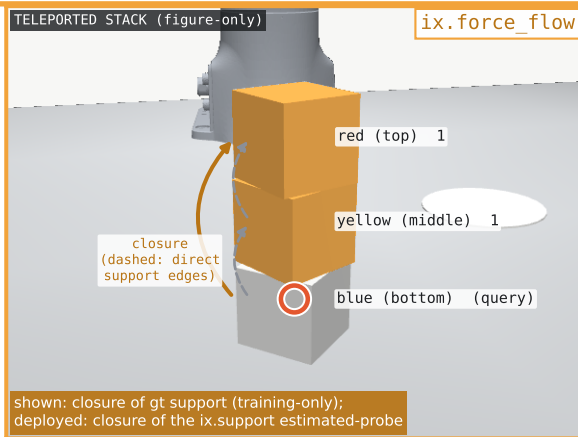


kin.sibling: dims at undirected tree distance 2: the sibling (r_shoulder_pitch) and, by the same op, grandparent and grandchild. hop x bias over the symmetrized kin_parent graph (hops=2; w=1). Source: public morphology (bodies.g1_simple).

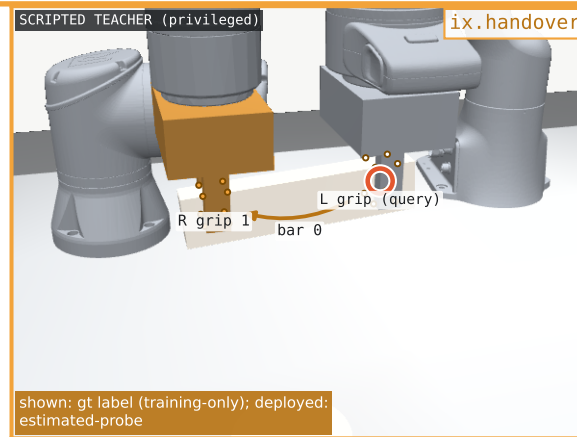
PHYSICAL INTERACTION (5 implemented)



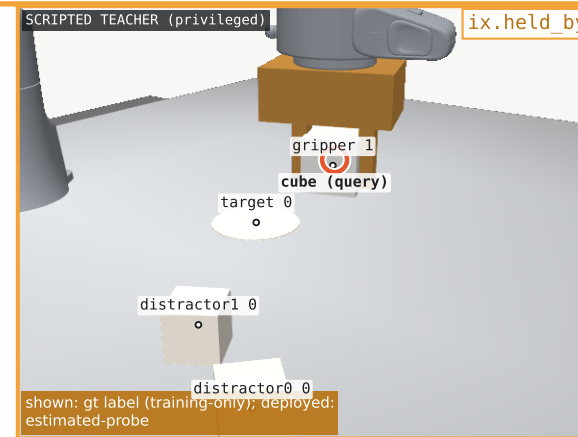
ix.contact: two entities in geometric contact (sim contact truth): query gripper -> cube 1, distractors / target 0. White dots: the gripper-cube contact points. bilinear x aug (probe). Source: shown: gt label (training-only); deployed: estimated-probe.



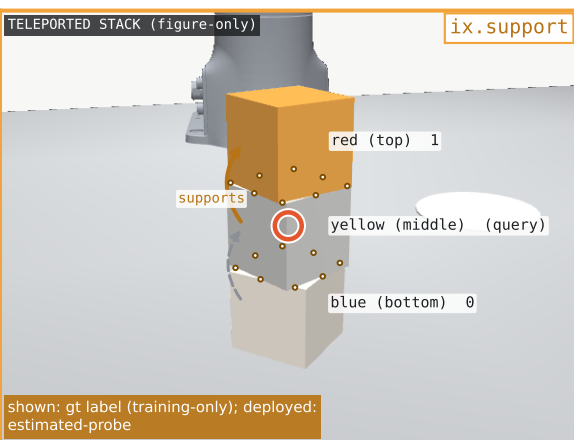
ix.force_flow: transitive closure of the support graph: everything the query carries, directly or through others. Bottom cube -> middle 1 and top 1 (direct edge only to the middle). Op: flow (closure) over support-v1. flow x bias. Source: shown: closure of gt support (training-only); deployed: closure of the ix.support estimated-probe.



ix.handover: two manipulator assemblies in contact with the same object at once (a handover in progress): query L grip -> R grip 1, bar / target 0. Dots: bar contacts of each hand. Dual env only. bilinear x aug (probe). Source: shown: gt label (training-only); deployed: estimated-probe.



ix.held_by: object held by a manipulator assembly (≥ 2 contacts with its hand bodies): lifted cube -> gripper 1; target and distractors 0. bilinear x aug (probe). Source: shown: gt label (training-only); deployed: estimated-probe.



ix.support: a supports b: contact + contact normal within 30° of gravity-up at a's top. Query middle cube -> top 1 (row); it is itself supported by the bottom cube (grey arrow, column). Emits the estimated support-v1 graph. bilinear x aug (probe, emits). Source: shown: gt label (training-only); deployed: estimated-probe.